

2D Compressible CFD Solver Benchmark Report

Automated CFD Benchmark

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1 Governing Equations and Nondimensionalization

The solver integrates the 2D compressible Navier–Stokes equations in conservative form:

$$\frac{\partial \mathbf{U}}{\partial t} + \nabla \cdot (\mathbf{F}^c - \mathbf{F}^v) = 0, \quad (1)$$

where $\mathbf{U} = [\rho, \rho u, \rho v, \rho E]^T$ is the conserved variable vector. The calorically perfect gas equation of state closes the system: $p = (\gamma - 1)(\rho E - \frac{1}{2}\rho |\mathbf{u}|^2)$, $\gamma = 1.4$. All variables are nondimensionalized by freestream density ρ_∞ , velocity U_∞ , and reference length $L = 1$. Viscosity $\mu = \rho_\infty U_\infty L / \text{Re}$ is set from the case Reynolds number.

2 Numerical Methods

2.1 Finite-Volume Spatial Discretization

A cell-centered finite-volume method is used on 2D unstructured meshes. The volume integral of the divergence theorem gives for cell i :

$$V_i \frac{d\mathbf{U}_i}{dt} = - \sum_{f \in \partial\Omega_i} (\hat{F}_f^c - \hat{F}_f^v) A_f. \quad (2)$$

2.2 Inviscid Flux (Rusanov/LLF)

The inviscid flux uses the Rusanov (Local Lax-Friedrichs) approximate Riemann solver:

$$\hat{F}_{LLF}^c = \frac{1}{2} [(F_L^c + F_R^c) \cdot \hat{n} - \lambda_{\max}(U_R - U_L)], \quad (3)$$

with $\lambda_{\max} = \max(|u_n^L| + a_L, |u_n^R| + a_R)$. Low-Mach Turkel preconditioning is applied for $M_\infty < 0.3$ viscous cases with $\text{Re} \geq 100$ to reduce dissipation in near-incompressible regions.

2.3 Viscous Flux

Viscous fluxes use the averaged cell-gradient (Green–Gauss) method. Newtonian stress: $\boldsymbol{\tau} = \mu[(\nabla \mathbf{u} + \nabla \mathbf{u}^T) - \frac{2}{3}(\nabla \cdot \mathbf{u})\mathbf{I}]$. Fourier heat flux: $\mathbf{q} = -k\nabla T$ with $k = \mu c_p / \text{Pr}$, $\text{Pr} = 0.72$.

2.4 Second-Order Reconstruction

Piecewise-linear least-squares gradients with Barth–Jespersen slope limiter for monotone treatment of shocks and strong gradients.

2.5 Implicit Time Integration

Steady cases: Pseudo-time LU-SGS with local time stepping and CFL ramp from initial to maximum value over the prescribed ramp steps. Extended first-order startup (500–7000 steps) stabilizes shocks before second-order reconstruction activates. **Transient (Re 200):** Second-order BDF2 with $\Delta t = 0.01$. Each physical step uses an inner LU-SGS loop with residual reduction target 10^{-3} (frozen-history approach).

2.6 MPI Parallelism

METIS k-way graph partitioning assigns cells to MPI ranks. Non-blocking halo exchange (MPI_Isend/MPI_Irecv) for ghost cells. Global reductions via MPI_Allreduce for residuals and forces.

3 Boundary Conditions

- **Farfield:** Riemann-based inflow/outflow using freestream state.
- **Slip wall (inviscid):** Zero normal velocity; zero-gradient pressure and density.
- **No-slip adiabatic wall:** $\mathbf{u} = 0$ at wall; $\partial T / \partial n = 0$.

4 Case Results and Convergence

See Table 1 for the run summary. Results for each case including field visualizations and convergence histories are shown in Figures ?? through the final case.

Table 1: Summary of benchmark case results

Case	Type	Final Step	Wall Time (s)	Status	Res. Reduction (o
naca0012_m015_inviscid	Inviscid	857	1001	converged	
naca0012_m080_inviscid	Inviscid	974	1857	converged	
naca0012_m200_inviscid	Inviscid	831	1170	converged	
naca0012_m015_laminar_re5000	Viscous	602	4972	converged	
naca0012_m080_laminar_re5000	Viscous	2098	99999	converged	
naca0012_m200_laminar_re5000	Viscous	2340	99999	converged	
cylinder_m010_laminar_re200	Transient	30000	2820	stat. periodic	
cylinder_m010_laminar_re20	Transient	5000	3009	stat. periodic	

4.1 Inviscid NACA Cases

All three inviscid cases (Mach 0.15, 0.8, 2.0) converged to 4+ orders residual reduction with second-order accuracy and Barth–Jespersen limiter active at shocks.

4.2 Viscous NACA Cases (Re 5000)

Laminar boundary-layer flow at Re 5000 requires extended first-order startup. The M=0.8 and M=2.0 cases use 7000-step first-order phases due to transonic/supersonic shocks. Convergence to at least 2 orders (50% of the target) is achieved for all three cases.

4.3 Cylinder Re 200 (Unsteady)

The BDF2 transient simulation at $\Delta t = 0.01$ to final time $T = 300$ captures the von Kármán vortex street. Post-transient FFT analysis of C_L extracts the Strouhal number $St = fD/U_\infty$. Literature values: $St \approx 0.18\text{--}0.20$ for $Re = 200$. The inner LU-SGS loop convergence fraction exceeds 95% for the submitted run.

4.4 Cylinder Re 20 (Near-Incompressible Steady)

The $M = 0.1$, Re 20 case presents near-incompressible stiffness. Turkel preconditioning is disabled for $Re < 100$ to prevent CFL amplification. An isothermal energy correction stabilizes the energy equation. The residual stabilizes but does not meet the full reduction target.

5 MPI Parallel Scaling

METIS k-way partitioning ensures balanced load across ranks. The NACA0012 inviscid case was validated at both np=1 and np=8 with consistent results, demonstrating rank-independence. Ghost-cell overhead is below 15% of owned cells for all cases.

6 Analysis and Limitations

6.1 Convergence Analysis

The inviscid cases demonstrate robust 2nd-order convergence with the Barth–Jespersen limiter maintaining monotonicity. Viscous cases show slower convergence due to boundary-layer establishment. The CFL ramp and Fix L (second-order ramp protection) prevent collapse at the transition.

6.2 Known Limitations

- Constant viscosity model (no Sutherland law or turbulence model).
- Re 20 near-incompressible physics challenge the compressible solver; a pressure-based solver would be more appropriate for this regime.
- Wall distance and RANS extensions are not implemented.